Direct rotation R

two principal planes

+45° −45°

Singular values of I − R

4 × 0.765…

trace sum = 3.061…

Competitor W

a moving and a fixed plane

+90° 0° (fixed)

Singular values of I − W

2 × 1.414… + 0 + 0

trace sum = 2.828…

Both maps send U to V; both principal angles are 45°.